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Bangabandhu Sheikh Mujibur Rahman Science and Technology University

Department of Computer Science and Engineering

4th Year 1st Semester B.Sc. Engineering Mid-term Examination-2023

Course Code: CSE407

Course Title: Digital System Design

Total Marks: 20

Time: 1 (One) Hour

Answer all questions.

1. a) Design an accumulator that can perform the following microoperations.

$p_1: A \leftarrow A + B$

$p_2: A \leftarrow A \vee B$

$p_3: A \leftarrow A \oplus B$

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- b) Determine the content of A after each microoperation if initially $A = 1100$ and $B = 1001$.

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2. Design an arithmetic logic circuit with three selection variables S_1, S_0 and C_{in} that generates the following operations. Draw the logic diagram of one typical stage.

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S_1	S_0	C_{in}	Function
0	0	0	$F = A + B$
0	0	1	$F = A$
0	1	0	$F = A + B + 1$
0	1	1	$F = A - 1$
1	0	0	$F = A + B' + 1$
1	0	1	$F = A + 1$
1	1	0	$F = A + B'$
1	1	1	$F = A$

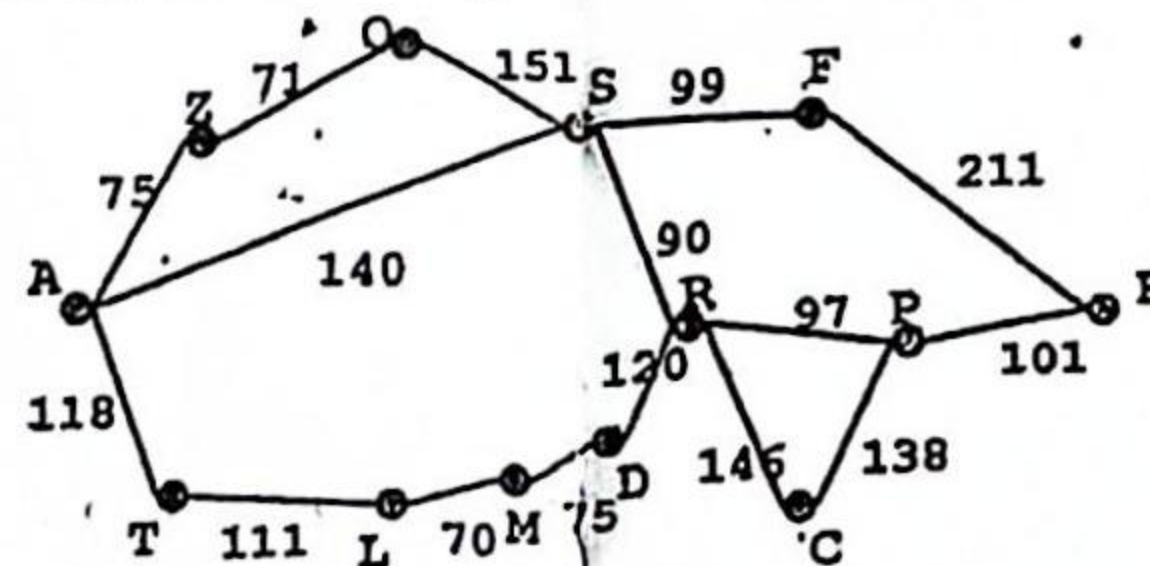
Bangabandhu Sheikh Mujibur Rahman Science and Technology University
Department of Computer Science and Engineering
4th Year 1st Semester B.Sc.Engg. Midterm Examination-2023

Course Code : CSE401
Course Title : Artificial Intelligence

Marks: 20
Time: 1 hour

N.B. Answer all 4(Four) questions

- Q1. What is an intelligent agent? Explain the significance of the statement - "System that acts rationally". 5
- Q2. How can you formally define a search problem? Explain using an example. "Uniform cost search acts like BFS when step costs are equal" - justify. 5
- Q3. Write short notes on the following terms: 5
(i) Uninformed Search
(ii) Episodic vs Sequential environment
- Q4. Find the shortest path measured by sum of distances along the path in the following graph using uniform cost search, where 'A' and 'B' are the initial and goal nodes respectively. 5



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Course No.: CSE411

Course Title: Communication Engineering

Full Marks: 20

Time: 1 Hours

N.B.: Answer any four (04) questions.

1. Define jitter. Write down about data flow. A sine wave is offset $1/5$ cycle with respect to time 0° . What is its phase in degrees and radians? 5
2. What is composite signal? What are the propagation time and the transmission time for a 2.5KB message if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12000 km and the light travels at 2.4×10^8 m/s. 5
3. Define baseband transmission and broadband transmission. What is the transmission time of a packet sent by a station if the length of the packet is 1 million bytes and the bandwidth of the channel is 200 Kbps? 5
4. What is transmission impairment? We have a channel with 4KHz bandwidth. If we want to send data at 100 Kbps, what is the minimum SNR_{db} ? What is the SNR? 5
5. Write down the factors of data rate. The signal-to-noise ratio is often given in decibels. Assume that $SNR_{db}=45$ and the channel bandwidth is 3 MHz. Calculate the channel capacity. 5

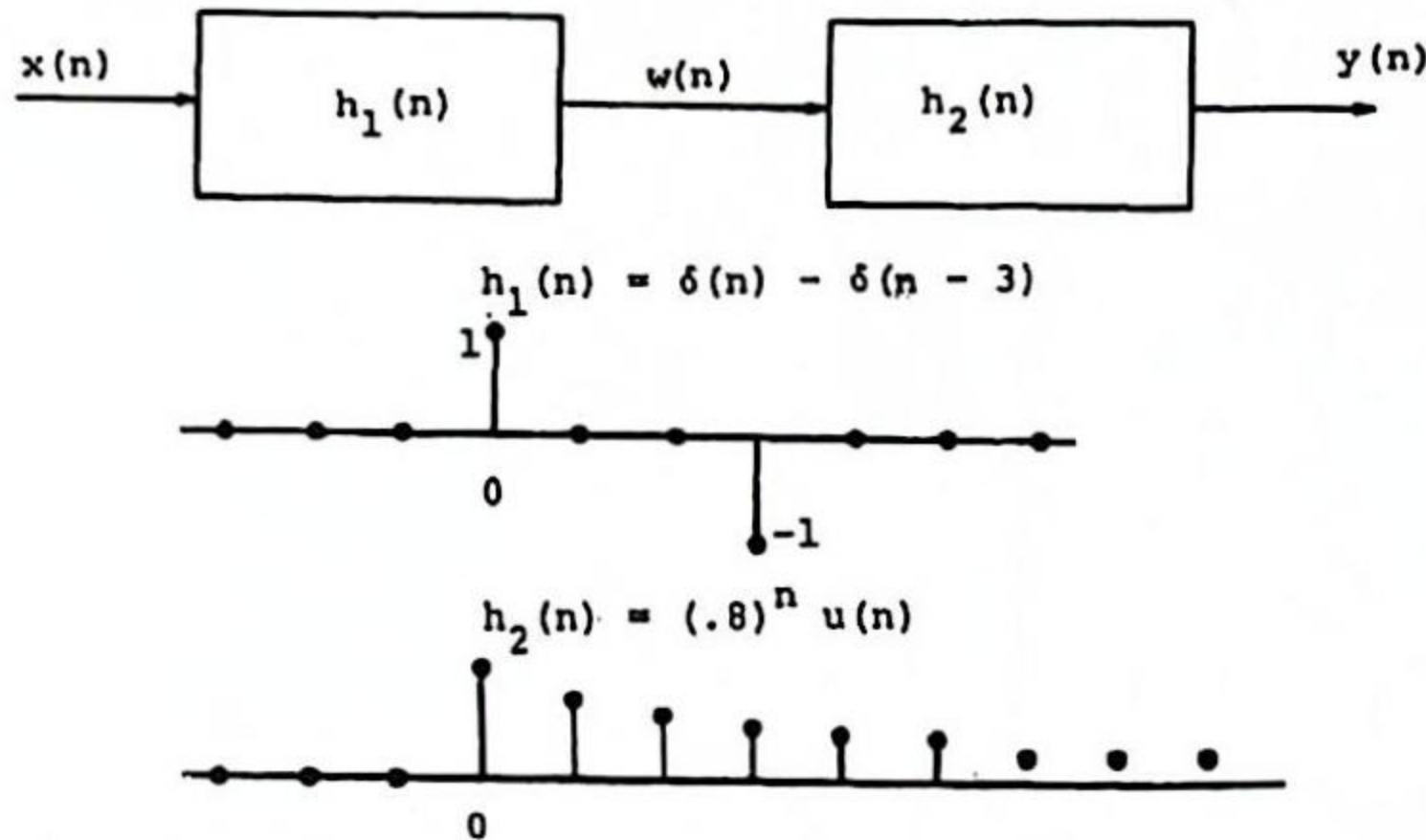
BNSNR

N.B.:

- 1) Answer any Four questions.
- 2) The figures in the right margin indicate full marks.

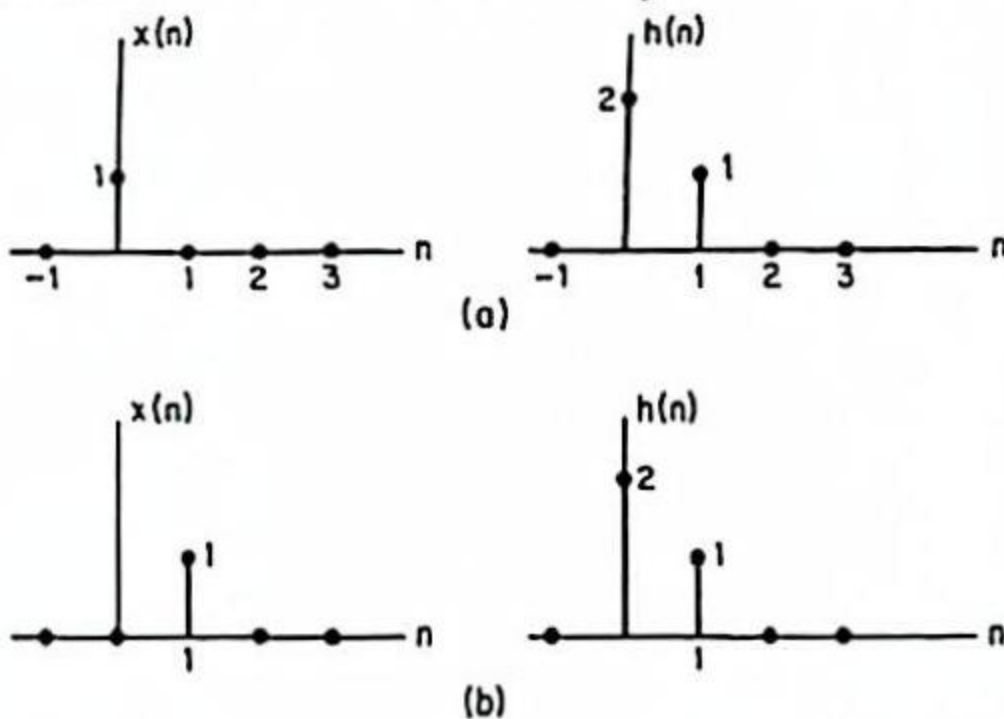
1. We can obtain the unit step sequence from the unit sample sequence and vice versa by constructing a running sum- Justify your answer. 5

2. 5



let $x(n) = u(n)$. Find $y_b(n)$ by convolving $x(n)$ with the result of the convolution of $h_1(n)$ and $h_2(n)$
i.e. $y_b(n) = x(n) * [h_1(n) * h_2(n)]$

3. For each of the sequences in below figure use discrete convolution to find the response to the input $x(n)$ of the linear shift-invariant system with unit-sample response $h(n)$. 5



- ✓ 4. Let $e(n)$ be an exponential sequence, i.e., $e(n) = \alpha^n$, for all n 5
and let $x(n)$ and $y(n)$ denote two arbitrary sequences. Show that

$$[e(n)x(n)] * [e(n)y(n)] = e(n)[x(n) * y(n)]$$

5. Consider a discrete-time linear shift-invariant system with unit-sample response $h(n)$. If the input $x(n)$ is a periodic sequence with period N , i.e., $x(n) = x(n + N)$, show that the output $y(n)$ is also a periodic sequence with period N . 5

Bangabandhu Sheikh Mujibur Rahman Science and Technology University

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4th Year 1st Semester B.Sc. Engg. Midterm Examination-2023 (Held on-2024) Session: 2019-20

Course No.: CSE403

Course Title: Computer Simulation and Modeling

Full Marks: 20

Time: 1 Hour

N.B. Answer any two of the following questions. Figures in the right-hand margin indicate full marks

1. a) What is meant by modeling and simulation? “Simulation is a valuable tool for many situations but not for all”-Justify this statement. 5
- b) Explain the steps followed in modelling? Write down the differences between deterministic and Stochastic models. 5
2. a) What is random number? List the properties of a good random number generator. 3
- b) How is Monte Carlo method different in approach from the typical mode of simulation, in deterministic problems? 4
- c) What is statistical distribution? Write down the measures of central tendency and measures of variation. 3
3. a) What is meant by continuous and discrete distribution. Also list the analysis steps involved in using normal distribution as a measurement model. 5
- b) Ada earned a score of 940 on a national achievement test. The mean test score was 850 with a standard deviation of 100. What proportion of students had a higher score than Ada? (Assume that test scores are normally distributed.) 5